

2019

Bachelor in Business Administration

Course Title: Financial Derivatives

Code No: **BNK205** Semester: **8th Semester**

Full Marks: 60 Pass Marks: 24 Time: of contract is 270.25. The initial margin per contract is \$2,500. And maintenance margin is 2,000. Settlement price on the first day is \$ 278.00 per ounce, second day is \$ 281.00 per ounce, third day is \$276.00, fourth day is \$280.00, and fifth day is \$275.00. Required: a. Calculate the daily gain or loss. b. Calculate the cumulative gain or loss. c. Calculate the margin account balance. d. Calculate the margin call. e. If investor does not deposit margin call, what will happen?

15. On a particular day, the September S&P; 500 stock index futures was priced at 960.50. The S&P; 500 index was at 956.49. The contract expires 73 days later. a. Assuming continuous compounding, suppose the risk-free rate is 5.96 percent, and the dividend yield on the index is 2.75 percent. Is the futures overpriced or underpriced? b. Assuming annual compounding, suppose the risk-free rate is 5.96 percent, and the future value of dividends on the index is \$5.27. Is the futures overpriced or underpriced?

16. Consider a plain vanilla interest rate swap with payments every 180 days (assume a 360-day year) for one year. The upcoming floating payment is at 5 percent. The notional principal is Rs 100 million. The prices of Eurodollar zero coupon bonds are as follows:

Day	Interest rate
180	5%
360	6%

a. Calculate the fixed rate on a plain vanilla interest rate swap. Why fixed rate is calculated? b. Calculate the value of plain vanilla swap from the perspective of the fixed rate payer. c. Calculate the fixed payment, floating payment, and net payment for the first period. [2+2+1]

Section "C" Comprehensive answer questions: [2 × 10 = 20]

17. The stock of Standard Chartered PLC is selling at Rs 500 per share and put option on this stock is available with the maturity period of one period from today. The strike price of a put is Rs 500 per share. It is expected that the price of stock one period from today will be either Rs 600 or Rs 400 per share. The risk free rate is 10 percent. a. What should be the price of put today? Why do you think it is the fair price of put? b. What do you mean by hedge portfolio? Can you create the hedge portfolio using puts and stocks? c. Show that the value of hedge portfolio will be same for both stocks prices at expiry. d. What should be the rate of return of hedge portfolio? Justify. e. Suppose the put is selling for Rs 20 per share. Suggest a strategy and calculate the profit at expiry.

18. The following option prices were observed for a stock for July 6 particular year. Ignore dividend and assume that the stock is priced at 165.13. The expirations are July 17, August 21, and October 16. The continuously compounded risk-free rates are 0.0503, 0.0535, and 0.0571, respectively. The standard deviation is 0.21. Assume that the options are European. Possible stock prices at the expiration date are Rs 155, Rs 160, Rs 165, Rs 170, Rs 175 and Rs 180.

Strike	Calls	Puts	July	Aug	Oct	July	Aug	Oct
165	2.705	2.58	102	40	170	803	256	005
170	7.57	509	00a					

a. Buy one August 165 call contract. Hold it until the option expires. Determine the profits and graph the results. Identify the breakeven stock price at expiration. What are the maximum possible gain and loss on this transaction? b. Buy one October 165 put contract. Hold it until the options expire. Determine the profits and graph the results. Identify the breakeven stock price at expiration. What are the maximum possible gain and loss on this transaction? c. Buy 100 shares of stock and write one October 165 call contract. Hold the position until expiration. Determine profits and graph the results. Identify the breakeven stock price at expiration, the maximum profit, and the maximum loss. d. Buy 100 shares of stock and buy one August 165 put contract. Hold the position until expiration. Determine profits and graph the results. Determine the breakeven stock price at expiration, the maximum profit, and the maximum loss.

Candidates are required to answer the questions in their own words as far as practicable.

Section A

Brief Answer Questions: [10 × 1 = 10]

1. Arbitrage is a transaction designed to capture profits resulting from market efficiency.

2. The bid price is the price paid to the buy an option from a market maker.
3. Buying a call with a lower exercise price offers a greater profit potential than one with a higher exercise price.
4. The maximum value of a call is the stock price.
5. Higher volatility in stock price increases the value of both call and put options.
6. One party to a futures transaction does not bear the risk that the other party will default.
7. Nepalese derivatives market is regulated by Securities Board of Nepal.
8. A generic asset that expires on September 10 whose spot price as of June 10 is Rs 40. Assume that the annual compounded risk free rate is 6 percent. The price of futures must be Rs 40.74870.
9. An interest rate swap is a special case of a currency swap with both currencies being the same.
10. Legal risk is the risk that the government will declare derivatives illegal.

Section B

Short Answer Questions:[6 × 5 = 30]

11. Describe the features of financial derivatives. Highlights the issues of derivatives market in Nepal.
12. From the following quotation, answer the questions given below.

Contract name	Last Trade	Put change	Bid	Ask	Vol	Interest	Strike price
MSFFU1V22925.01	630.071	621.644	043,856	25a.			

Describe the meaning of terms of column first.
 b. What was the price of a put on the quotation day and on day before quotation?
 c. What does 'Interest' mean?
 d. What price an investor should pay to buy a put?
 e. How many options have been traded on quotation day? What is bid-ask spread?
13. Consider a stock of Lama Corporation (LC) that trades for Rs 100 per share. Call and put options on this stock have an exercise price of Rs 100 and they expire in 1 year. The risk-free rate is 12 percent per annum, and the standard deviation of the stock return is 16 percent. Determine the value of put option on the LC's stock by using Black and Scholes Model. If the current value of the put option on the LC's stock is Rs 2, what should an investor should decide? $[W=e^{-H}]$
14. A company enters into a short futures position in 10 contracts of gold at a futures price of \$ 276.50 per ounce. Each contract is for 100 ounces. Spot price at the time of contract is 270.25. The initial margin per contract is \$2,500. And maintenance margin is 2,000. Settlement price on the first day is \$ 278.00 per ounce, second day is \$ 281.00 per ounce, third day is \$276.00, fourth day is \$280.00, and fifth day is \$275.00. Required:
 a. Calculate the daily gain or loss.
 b. Calculate the cumulative gain or loss.
 c. Calculate the margin account balance.
 d. Calculate the margin call.
 e. If investor does not deposit margin call, what will happen?
15. On a particular day, the September S&P; 500 stock index futures was priced at 960.50. The S&P; 500 index was at 956.49. The contract expires 73 days later.
 a. Assuming continuous compounding, suppose the risk-free rate is 5.96 percent, and the dividend yield on the index is 2.75 percent. Is the futures overpriced or underpriced?
 b. Assuming annual compounding, suppose the risk-free rate is 5.96 percent, and the future value of dividends on the index is \$5.27. Is the futures overpriced or underpriced?
16. Consider a plain vanilla interest rate swap with payments every 180 days (assume a 360-day year) for one year. The upcoming floating payment is at 5 percent. The notional principal is Rs 100 million. The prices of Eurodollar zero coupon bonds are as follows:

Day	Interest rate	1805%	3606%
a.			

Calculate the fixed rate on a plain vanilla interest rate swap. Why fixed rate is calculated?
 b. Calculate the value of plain vanilla swap from the perspective of the fixed rate payer.
 c. Calculate the fixed payment, floating payment, and net payment for the first period. $[2+2+1]$

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- What should be the price of put today? Why do you think it is the fair price of put?
 - What do you mean by hedge portfolio? Can you create the hedge portfolio using puts and stocks?
 - Show that the value of hedge portfolio will be same for both stocks prices at expiry.
 - What should be the rate of return of hedge portfolio? Justify.
 - Suppose the put is selling for Rs 20 per share. Suggest a strategy and calculate the profit at expiry.
18. The following option prices were observed for a stock for July 6 particular year. Ignore dividend and assume that the stock is priced at 165.
- | Strike | Calls | Puts |
|----------|-------|------|
| July 165 | 2.70 | 5.25 |
| Aug 165 | 5.10 | 2.40 |
| Oct 165 | 7.80 | 0.25 |
- The expirations are July 17, August 21, and October 16. The continuously compounded risk-free rates are 0.0503, 0.0535, and 0.0571, respectively. The standard deviation is 0.21. Assume that the options are European. Possible stock prices at the expiration date are Rs 155, Rs 160, Rs 165, Rs 170, Rs 175 and Rs 180.
- Buy one August 165 call contract. Hold it until the option expires. Determine the profits and graph the results. Identify the breakeven stock price at expiration. What are the maximum possible gain and loss on this transaction?
 - Buy one October 165 put contract. Hold it until the options expire. Determine the profits and graph the results. Identify the breakeven stock price at expiration. What are the maximum possible gain and loss on this transaction?
 - Buy 100 shares of stock and write one October 165 call contract. Hold the position until expiration. Determine profits and graph the results. Identify the breakeven stock price at expiration, the maximum profit, and the maximum loss.
 - Buy 100 shares of stock and buy one August 165 put contract. Hold the position until expiration. Determine profits and graph the results. Determine the breakeven stock price at expiration, the maximum profit, and the maximum loss.